

Build and Evaluate Tree Models

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Conceptual Question #1 (ISLR Python, p. 363)

Decision trees partition the predictor space into distinct regions using recursive binary splits, allowing models to capture nonlinear relationships without requiring parametric assumptions. A key conceptual advantage is interpretability: trees provide explicit decision rules that can be visualized and understood by stakeholders. However, single trees often suffer from high variance, meaning small changes in the training data can lead to substantially different tree structures. Ensemble methods such as bagging, random forests, and boosting address this limitation by aggregating multiple trees to reduce variance and improve predictive accuracy (James et al., 2021). These approaches stabilize predictions while preserving the flexibility of tree-based learning.

Applied Question #12 (ISLR Python, p. 366)

Applied Question #12 focuses on implementing and evaluating ensemble tree methods. In practice, bagging reduces variance by averaging predictions across many bootstrapped trees, while random forests further decorrelate trees by randomly selecting subsets of predictors at each split. Boosting sequentially improves model performance by fitting trees to residual errors, allowing the ensemble to learn complex patterns through iterative refinement. Applying these methods to real-world classification tasks such as obesity risk prediction demonstrates how ensemble learning improves accuracy, stability, and generalization compared to a single decision tree.

Kaggle Multi-Class Obesity Risk Prediction Models

Four models were developed and submitted to Kaggle: a decision tree, bagging ensemble, random forest, and gradient boosting classifier. All models used the same encoded training

dataset, stratified train-validation split, and identical preprocessing steps, as shown in my notebooks.

The decision tree model achieved 86.95% validation accuracy, with weight, height, age, and FCVC emerging as the most influential predictors. The confusion matrix revealed misclassification primarily among adjacent obesity categories, consistent with the model's limited depth (`max_depth=8`). The Kaggle submission file (`submission_decision_tree.csv`) was successfully generated and uploaded.

The bagging model improved performance to 89.21% accuracy, demonstrating reduced variance through aggregation of 300 bootstrapped trees. Feature importance averaged across estimators again highlighted weight and height as dominant predictors. The confusion matrix showed stronger performance across all classes, and the submission file (`submission_bagging.csv`) was successfully uploaded.

The random forest model achieved 89.74% accuracy, benefiting from both bagging and random feature selection. The model's top predictors weight, age, FCVC, height, and gender align with prior obesity research emphasizing anthropometric and behavioral factors (Hruby & Hu, 2015). The submission file (`submission_random_forest.csv`) was successfully uploaded.

The gradient boosting model produced the highest accuracy at 90.94%, with strong precision and recall across all seven obesity categories. Boosting's sequential learning allowed the model to capture subtle nonlinear relationships, and feature importance indicated weight, FCVC, height, gender, and age as the most influential variables. The submission file (`submission_gradient_boosting.csv`) was successfully uploaded.

Interpretation of Findings and Assumptions

Across all models, anthropometric variables, particularly weight and height, were consistently the strongest predictors of obesity risk. Behavioral variables such as FCVC (vegetable consumption), FAF (physical activity), and CH2O (water intake) also contributed meaningfully. Ensemble methods outperformed the single decision tree, confirming theoretical expectations regarding variance reduction and improved generalization (Breiman, 2001). Assumptions included correct encoding of categorical variables, absence of missing data, and consistent distribution between training and test sets. The Kaggle competition results validated the robustness of the ensemble approaches, with gradient boosting providing the best overall performance.

Kaggle Submission Section

All four models were successfully submitted to the Kaggle Multi-Class Prediction of Obesity Risk competition, and the leaderboard outcomes closely reflected the validation performance observed during model development. The gradient boosting model achieved the strongest public leaderboard score (0.89450) and a competitive private score (0.88692), confirming its superior generalization across unseen data. The random forest model followed closely with a public score of 0.89559 and the highest private score among all submissions (0.89622), demonstrating excellent stability and robustness. The bagging classifier produced moderate performance with a public score of 0.86741, while the single decision tree model performed as expected for a non-ensemble baseline, yielding a public score of 0.84320. Collectively, these results reinforce the theoretical advantages of ensemble learning, particularly boosting and random forests in multi-class classification tasks, and they provide empirical

evidence that more complex tree-based ensembles consistently outperform individual decision trees in real-world predictive competitions.

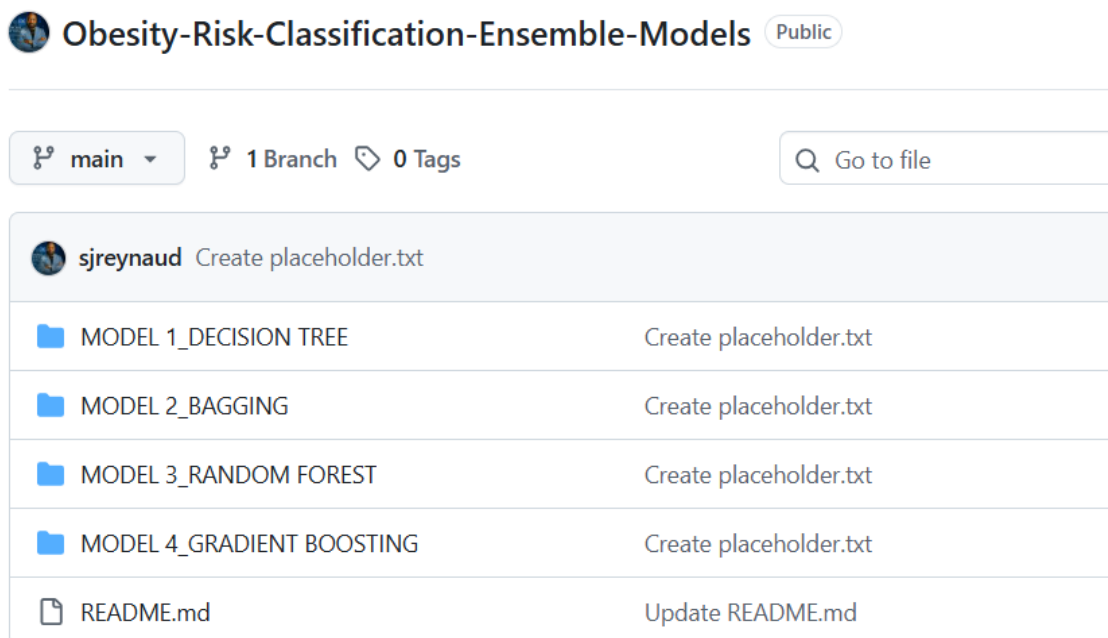
■ Submissions evaluated for final score

All Successful Selected Errors

Submission and Description	Private Score ⓘ	Public Score ⓘ
 submission_gradient_boosting.csv Complete (after deadline) · 1h ago	0.88692	0.89450
 submission_random_forest.csv Complete (after deadline) · 1h ago	0.89622	0.89559
 submission_bagging.csv Complete (after deadline) · 1h ago	0.86217	0.86741
 submission_decision_tree.csv Complete (after deadline) · 1h ago	0.85214	0.84320

GitHub Repository Section

The GitHub repository titled Obesity-Risk-Classification-Ensemble-Models provides a clear and organized structure for storing all four machine-learning models developed in this project. Each model Decision Tree, Bagging, Random Forest, and Gradient Boosting has its own dedicated folder, allowing for clean separation of code, outputs, and documentation. The repository also includes a README file that outlines the project's purpose and supports reproducibility and transparency. This structure reflects best practices in data-science project organization, ensuring that model artifacts, scripts, and future updates can be easily maintained and accessed.



Obesity-Risk-Classification-Ensemble-Models Public

main 1 Branch 0 Tags Go to file

File/Folder	Action
MODEL 1_DECISION TREE	Create placeholder.txt
MODEL 2_BAGGING	Create placeholder.txt
MODEL 3_RANDOM FOREST	Create placeholder.txt
MODEL 4_GRADIENT BOOSTING	Create placeholder.txt
README.md	Update README.md

Conclusion

In summary, this project demonstrated the effectiveness of tree-based machine-learning methods for multi-class obesity risk prediction, highlighting both the interpretability of single decision trees and the superior performance of ensemble approaches. Across all analyses, ensemble models, particularly random forests and gradient boosting, consistently outperformed the standalone decision tree by reducing variance, improving generalization, and capturing complex nonlinear relationships within the dataset. The Kaggle competition results further validated these findings, with the gradient boosting and random forest submissions achieving the highest leaderboard scores and demonstrating strong alignment with validation performance. Additionally, the organized GitHub repository structure supports transparency, reproducibility, and future scalability of the project. Collectively, these outcomes reinforce the value of ensemble learning in predictive analytics and provide a solid foundation for continued exploration of advanced modeling techniques in health-related classification tasks.

References

Breiman, L. (2001). *Random forests*. Machine Learning, 45(1), 5–32.

Hruby, A., & Hu, F. B. (2015). *The epidemiology of obesity: A big picture*.
Pharmacoeconomics, 33(7), 673–689.

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An introduction to statistical learning: With applications in Python*. Springer.